University of Batna2
Computer Science Department

4 th year engineer SC
Operating System Security

PW °3

Attacks on Operating System Security

Section 1: Reconnaissance and Information Gathering Attacks

1.1 Passive Reconnaissance

```
└─$ whois univ-batna2.dz
% This is the WHOIS server for Example
Registry.
% Queries are subject to terms of use.
%
Domain Name:     UNIV-BATNA2.DZ
Registrar:       wissal
Registrant:      Universite Batna 2
Adm Contact:     Dekhinet Abdelhamid
Adm Email:       a.dekhinet@gmail.com
Tech contact:    Dekhinet Abdelhamid
Tech Email:      a.dekhinet@gmail.com
Creation Date:   2016-03-22
Expiry Date:     N/A

% Query served at 2026-04-21T10:53:27.6
54262091+01:00
```

we extraxt a lot of inmportant info
like the admin of the website is dekkent and we can performe phising attack againset him

now after we know the Owner and the admin of the website let's extract more important info like the IPs of the servers and also extraxt the hiden servers

```
└─$ dig univ-batna2.dz

; <<>> DiG 9.20.22-1-Debian <<>> univ-b
atna2.dz
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status:
NOERROR, id: 5125
;; flags: qr rd ra; QUERY: 1, ANSWER: 1
, AUTHORITY: 0, ADDITIONAL: 1
```

nice we get that the domine related with this ip

```
;; ANSWER SECTION:
univ-batna2.dz.          9929    IN    A
41.111.218.138
```

also it's tell us that the TTL = 9929 wich mean that any packet will store in the cahe for 2.5 houre

now let 's search for the mail server and we will try to attack it later using hydra
the number(1,5,10) means the prorety it send it to 1 and if it fail it send it to 5

```
└─$ dig univ-batna2.dz MX

; <<>> DiG 9.20.22-1-Debian <<>> univ-b
atna2.dz MX
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status:
NOERROR, id: 41226
;; flags: qr rd ra; QUERY: 1, ANSWER: 5
, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1220
; COOKIE: 1967281cf0f551ba0100000069e74
```

very good we find that the unversity use google workspace for manging the mail server
now we need to change the startgie because it's impossible to run brutefoce attack against google mail server it will block us imddietly

```
univ-batna2.dz.          18000    IN     M
X        1 ASPMX.L.GOOGLE.COM.
```

now let's serach for any leaked data or info of the internt using theharveter tool

i try 3 defreet search engine and i get nothong *_*

```
└─$ theHarvester -d univ-batna2.dz -b g
oogle
Read proxies.yaml from /etc/theHarveste
r/proxies.yaml
*******************************************
*****************************
```

```
*
                                        *
*******************************************
***************************
The following engines are not supported
: {'google'}

[!] Invalid source.
```

```
└─$ theHarvester -d univ-batna2.dz -b b
aidu
Read proxies.yaml from /etc/theHarveste
r/proxies.yaml
*******************************************
***************************
```

so i try all the search engine possible and i didn't set any APK key and iget very interesting thing

```
└─$ theHarvester -d univ-batna2.dz -b a
ll
Read proxies.yaml from /etc/theHarveste
r/proxies.yaml
*******************************************
***************************
```

i found other IP next with the privousess one 41.11.218.138

```
[*] IPs found: 2
--------------------
193.194.68.230
41.111.218.138
```

and also all the sub domine of the university like bio math ….

```
activite.univ-batna2.dz:rproxy.univ-bat
na2.dz
aps.univ-batna2.dz:rproxy.univ-batna2.d
z
apsa.univ-batna2.dz:rproxy.univ-batna2.
dz
bac.univ-batna2.dz:rproxy.univ-batna2.d
z
bac.univ-batna2.dz:ww.univ-batna2.dz.
bibliotheque.univ-batna2.dz:rproxy.univ
```

and a lot of emails

```
-----------------------
azeddine.djennane@univ-batna2.dz
dep.gm@univ-batna2.dz
dep.medecine@univ-batna2.dz
pedagogie.scmi@univ-batna2.dz
s.benharzallah@univ-batna2.dz
s.yahiaoui@univ-batna2.dz
sg@univ-batna2.dz
st@univ-batna2.dz
vrpg@univ-batna2.dz
```

now we want to draw the infrastracter of the hole system (networek , serer ,proxy ,WAF….)

1.2 Active Reconnaissance

1) chek for firewall

we have this nmap commande that show if there is firewall or no
we can simpply chek for firewall existence by doing this nmap commad

```
┌──(ababil㉿AALIIWAAMRA)-[~]
└─$ sudo nmap -sA 41.111.218.138
```

this commande worek like that
send tcp with ACK flag to the given IP with no context (pre connection)
if the firewall interspet the packet he will ignore the packet or will
response with ICMP error

and if a host recive the packet wich means no firewall he will reponse with RST

```
Host is up (0.38s latency).
All 1000 scanned ports on 41.111.218.138 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 108.77 seconds
```

so from the this output we figure out that there is a statfull firewall

nice now let's try to do a service nmap scan to se the vernalle service

```
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
24/tcp    open  priv-mail
25/tcp    open  smtp
26/tcp    open  rsftp
30/tcp    open  unknown
32/tcp    open  unknown
33/tcp    open  dsp
37/tcp    open  time
42/tcp    open  nameserver
43/tcp    open  whois
49/tcp    open  tacacs
53/tcp    open  domain
70/tcp    open  gopher
79/tcp    open  finger
```

hmmmmm it sems like a honypot or something we need to cheek it out (^__^)

also it show a known back door port like Doom (666) , Elite (31337) , Netbus (12345)

so after run this nmap scan i notice that there something block my IP so i try to spoof my IP using TOR proxy and this make me think that there is a IDS/IPS instead of HOnypoet but am still not sure

scan without my real IP

```
┌──(ababil㉿AALIIWAAMRA)-[~]
└─$ sudo nmap -sV -p 21,445,3389
,31337 41.111.218.138
Starting Nmap 7.99 ( https://nma
p.org ) at 2026-04-21 23:02 +010
0
Stats: 0:00:12 elapsed; 0 hosts
completed (1 up), 1 undergoing S
ervice Scan
Service scan Timing: About 0.00%
 done
Stats: 0:00:18 elapsed; 0 hosts
completed (1 up), 1 undergoing S
ervice Scan
```

scan after using Tor proxy

```
┌──(ababil㉿AALIIWAAMRA)-[~]
└─$ sudo proxychains nmap -sV -p
 21,445,3389,31337 41.111.218.13
8
```

```
PORT      STATE SERVICE    VERSI
ON
21/tcp    open  tcpwrapped
445/tcp   open  tcpwrapped
3389/tcp  open  tcpwrapped
31337/tcp open  tcpwrapped
```

hmmmmm that make the assumption of there is a Honypoet more logique
because we got the in the name of the service tcpwrapped wich mean when 3-way hand shak happen and the nmap try to request the servie version the (IPS/Honypoet) close the connection

```
┌──(ababil㉿AALIIWAAMRA)-[~]
└─$ sudo proxychains nmap -sS -sV -p 22,80 41.111.218.138
[proxychains] config file found: /etc/proxychains4.conf
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4
[proxychains] DLL init: proxychains-ng 4.17
[proxychains] DLL init: proxychains-ng 4.17
[proxychains] DLL init: proxychains-ng 4.17
[proxychains] DLL init: proxychains-ng 4.17
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-21 23:29 +0100
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  41.111.218.138:22 <--socket error or timeout!
Stats: 0:00:16 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 50.00% done; ETC: 23:29 (0:00:00 remaining)
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  41.111.218.138:80  ...  OK
Stats: 0:00:22 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 50.00% done; ETC: 23:29 (0:00:22 remaining)
Stats: 0:00:22 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 100.00% done; ETC: 23:29 (0:00:00 remaining)
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  41.111.218.138:80  ...  OK
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  41.111.218.138:80  ...  OK
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  41.111.218.138:22 <--socket error or timeout!
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  41.111.218.138:80  ...  OK
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  41.111.218.138:80  ...  OK
Stats: 0:00:38 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 95.65% done; ETC: 23:29 (0:00:01 remaining)
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  41.111.218.138:80  ...  OK
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  41.111.218.138:80  ...  OK
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  41.111.218.138:80  ...  OK
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  41.111.218.138:80  ...  OK
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  41.111.218.138:80  ...  OK
Nmap scan report for 41.111.218.138
Host is up (0.14s latency).

PORT   STATE SERVICE    VERSION
22/tcp open  tcpwrapped
80/tcp open  http       OpenResty web app server

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 48.23 seconds
```

after this scan the image get more clear there are :

- OpenResty act like edge Getway and play the role of :
  - reverse proxy receive the request from the port 80 and 443 and forwarded it to internal servers
  - WAF scan the content of the HTTP request

- IPS withe decaption feature (act like a honypot) : instead of cut the connaction it send you to a service like a honypoet to wast tour time

approximate infrastructure :

Internet --> OpenResty (WAF/IPS/Proxy) --> Internal Network (172.16.0.x)

Section 2: Password Attacks